SIGHT: AI-Driven Immersive Safety Training for Industry 5.0 Manufacturing

Dr. Arthur Carvalho
arthur.carvalho@miamioh.edu

MIAMI UNIVERSITY

Manufacturing is a cornerstone of the U.S. economy

$2.9 Trillion

Annual contribution to U.S. GDP.

12.7 Million

Workers employed in the sector.

54%

Share of R&D conducted by American businesses.

Yet, persistent safety challenges cary a severe human and financial cost

>320,000
Workplace injuries reported annually

$7.5 Billion
Estimated annual direct costs (medical payments and lost wages)

11 Days
Average lost workdays per non-fatal injury

Source: U.S. Bureau of Labor Statistics (2023), Liberty Mutual Insurance (2025)

Manufacturing

- Contributing factor
  - Manufacturing training is challenging

  - Trainees must:
    - Review machine manuals
    - Watch tutorials and demos
    - Listen to instructors

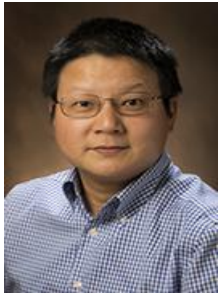
SIGHT

- **S**afety **I**mmersion and **G**amified **H**azard **T**raining (**SIGHT**)
  - $1.5 million grant from the Ohio Bureau of Workers' Compensation (BWC) through its Worker Safety Innovation Center (WSIC)

Dr. Fadel Megahed
Role: PI

Dr. Arthur Carvalho
Role: Co-PI

Dr. Jay Shan
Role: Co-PI



Dr. Mohammad Mayyas
Role: Co-PI

Dr. Reza Abrishambaf
Role: Sr. Personnel

Dr. Ibrahim Yousif
Role: Postdoc

Department of Information Systems and Analytics

Department of Engineering Technology

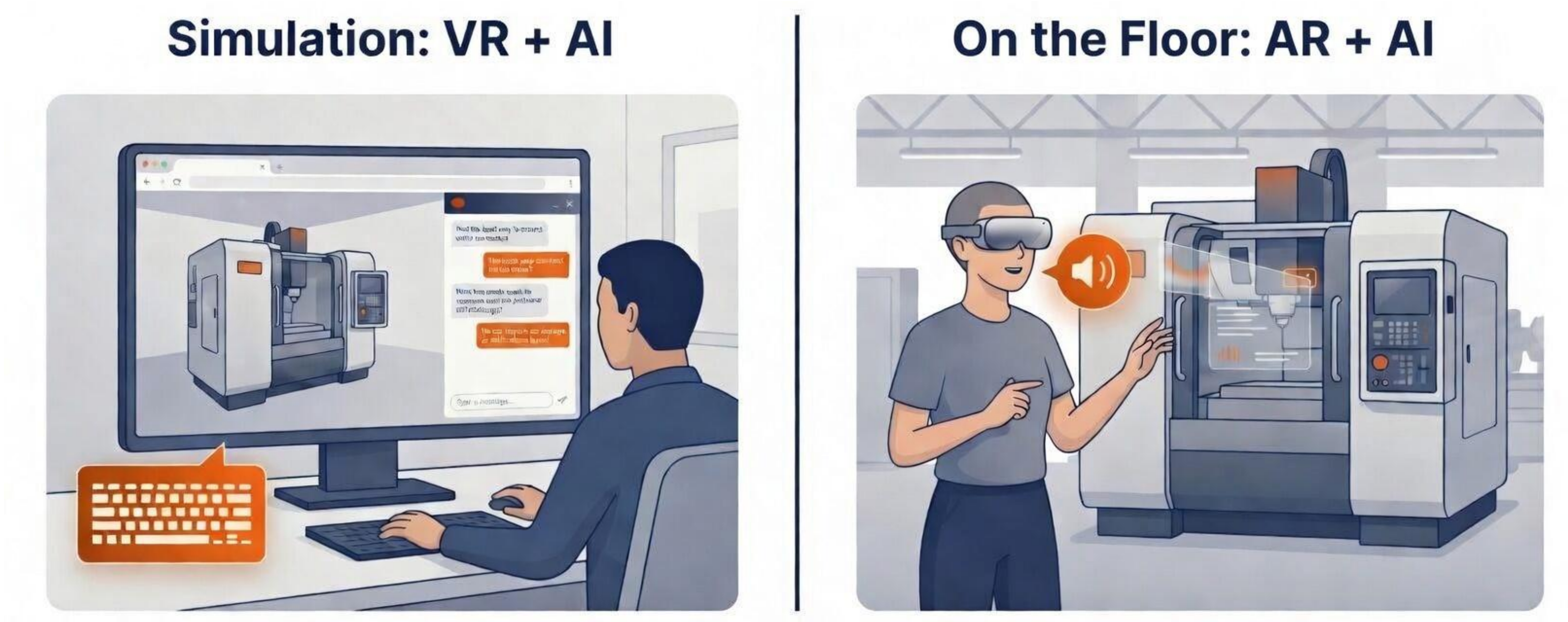
meetkai

maxbyte.
inspiring industry x.0

MIAMI UNIVERSITY

SIGHT

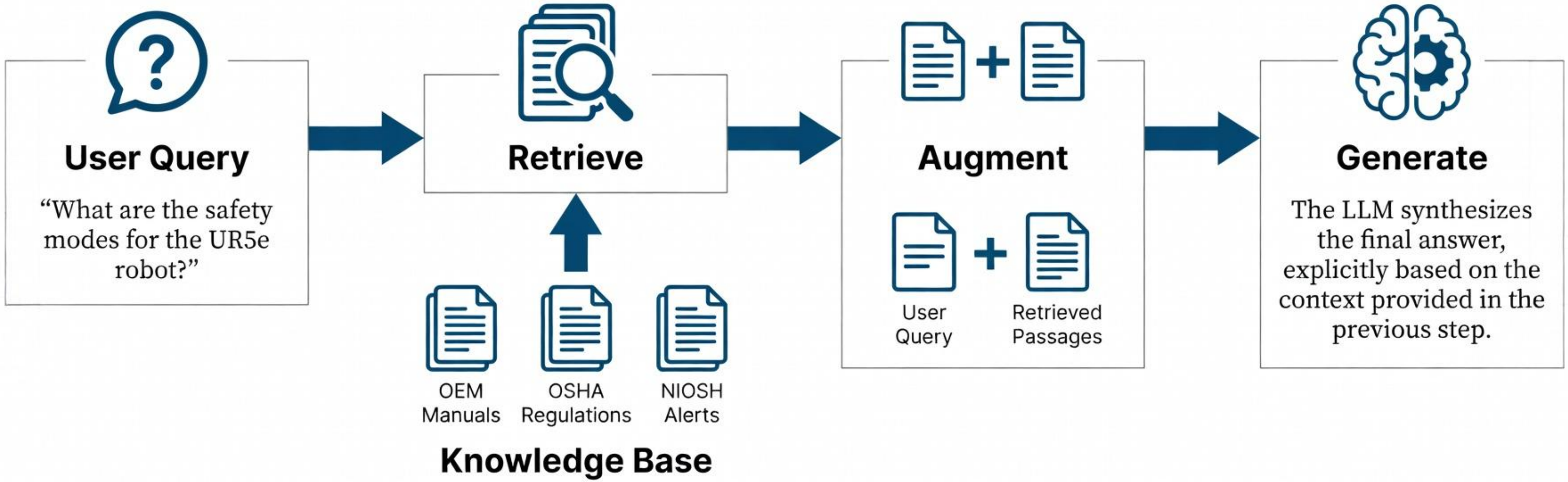
- Two manufacturing training solutions:



MIAMI UNIVERSITY

AI Component

- **R**etrieval **A**ugmented **G**eneration (RAG) technique



Experiments

- RAG pipelines

Factor	Levels
Retrieval Method	OpenAI Keyword OpenAI Semantic BM25 Graph Eager Graph MMR Vanilla Similarity
Large Language Model	gpt-5-mini-2025-08-07 gpt-5-nano-2025-08-07
Retrieval Depth	Top 3 Chunks (k=3) Top 7 Chunks (k=7)

Total Configurations Tested:
$6 \times 2 \times 2 = 24$ Pipelines

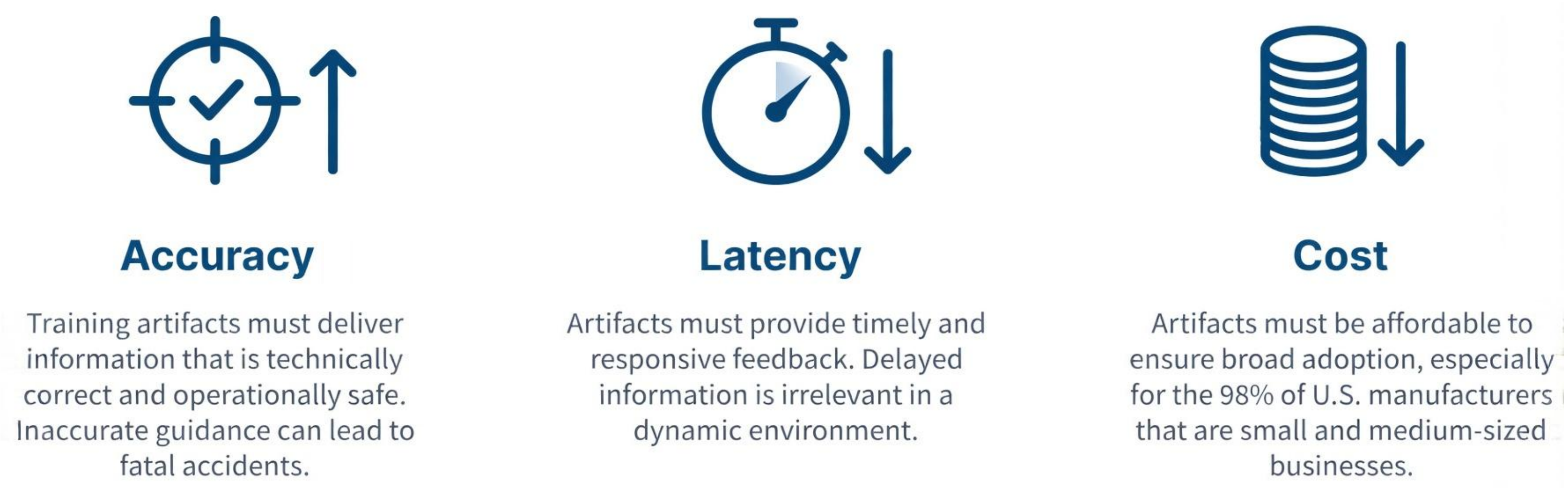
Experiments

- Comprehensive benchmark of Q&A pairs
  - Derived from official operational and safety manuals

  - Our results focus on the UR5e Cobot

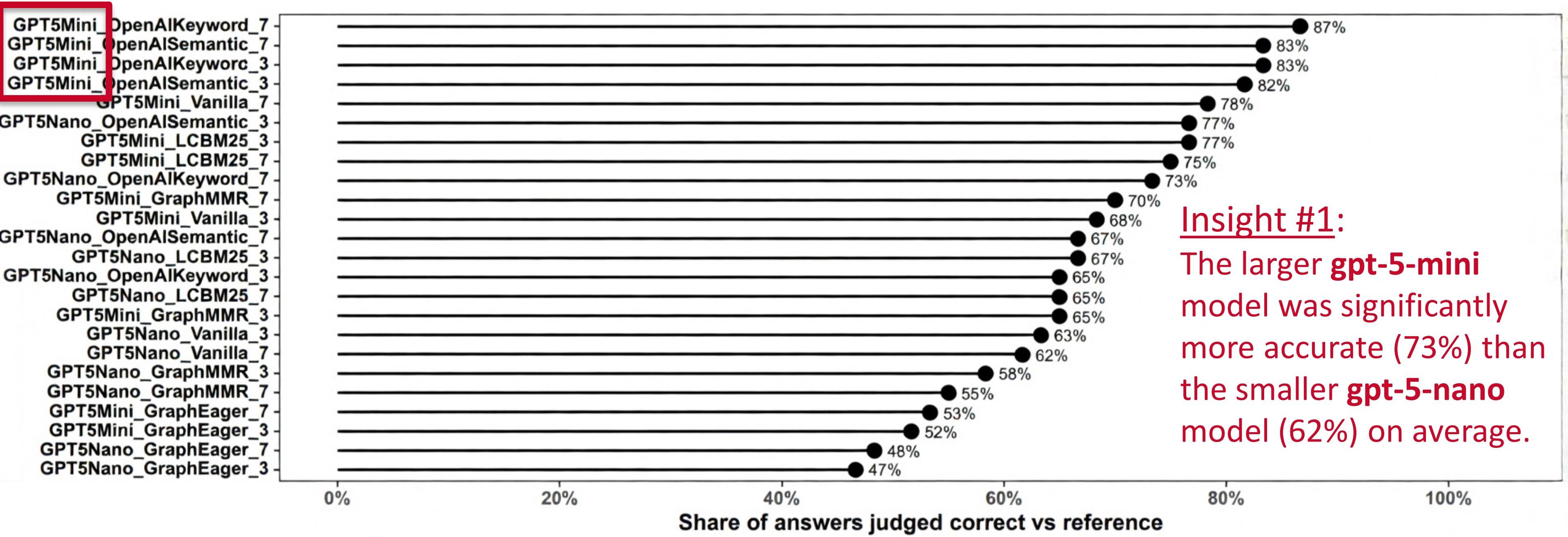
Experiments

- Three metrics



Results

➢ **Accuracy**: Model size and retrieval strategies are critical

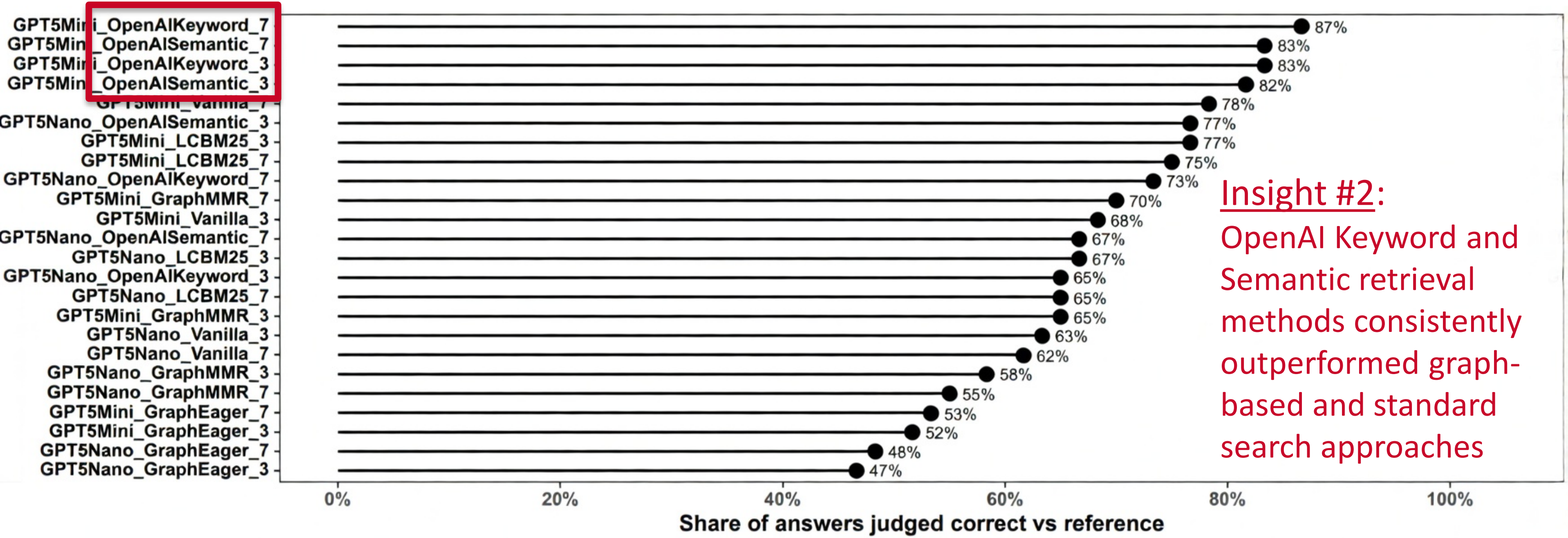


Insight #1:
The larger **gpt-5-mini** model was significantly more accurate (73%) than the smaller **gpt-5-nano** model (62%) on average.

MIAMI UNIVERSITY

Results

➢ **Accuracy**: Model size and retrieval strategies are critical

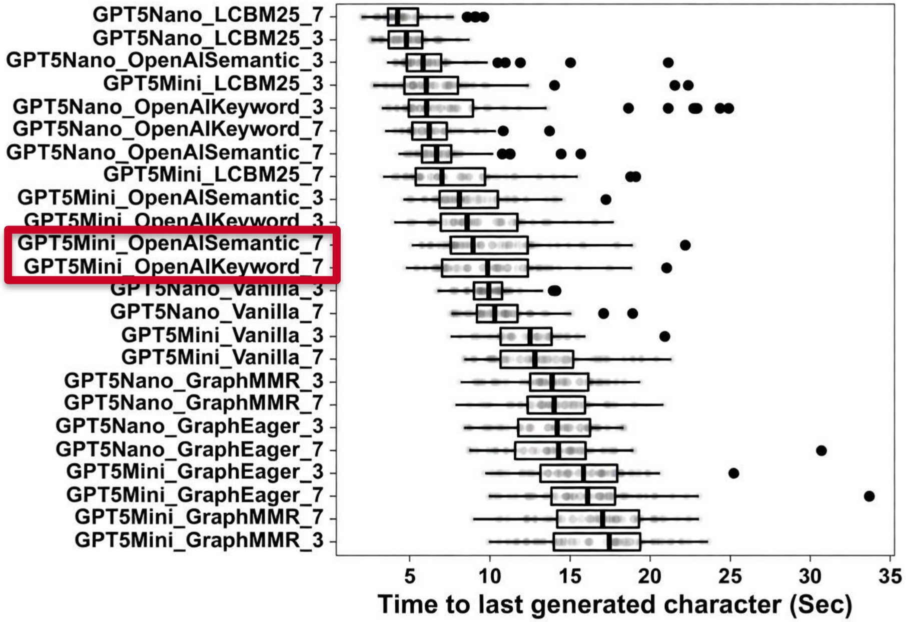


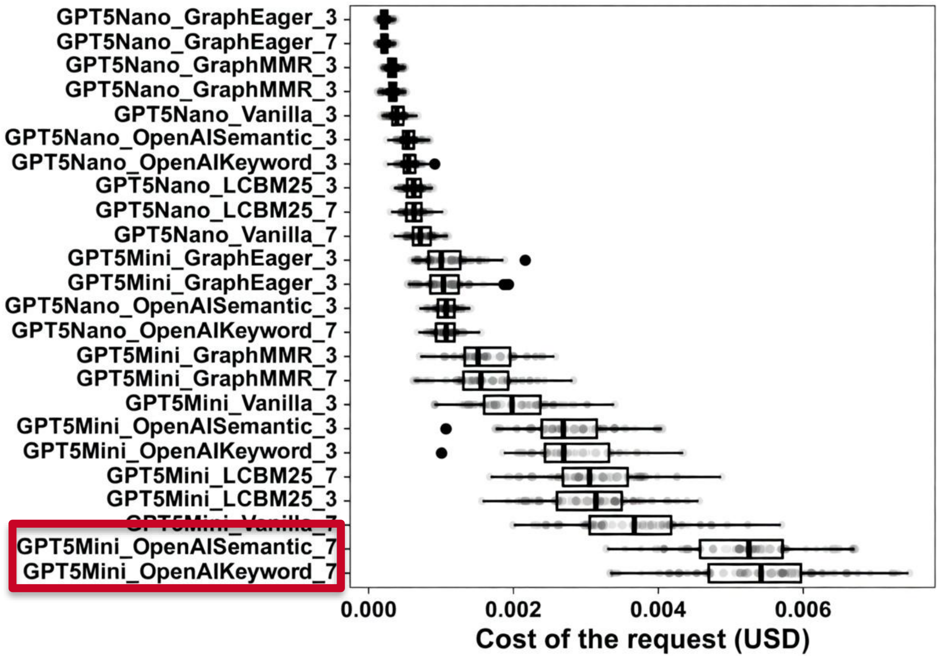
Insight #2:
OpenAI Keyword and Semantic retrieval methods consistently outperformed graph-based and standard search approaches

MIAMI UNIVERSITY

Results

➢ **<u>Latency</u>** and **<u>Cost</u>**: Trade-offs

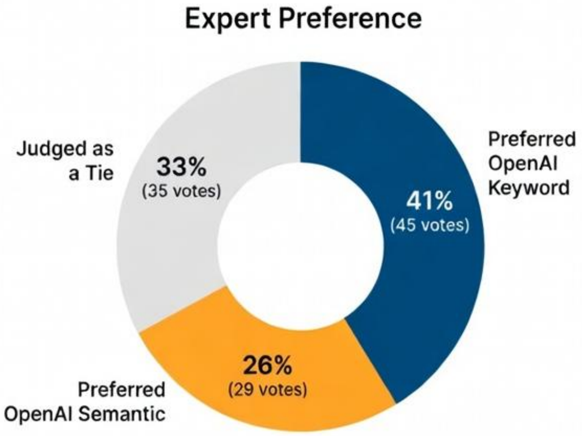


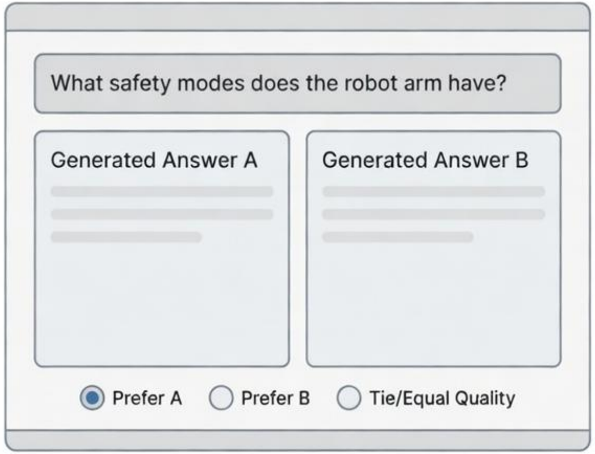


MIAMI UNIVERSITY

Results

- No single pipeline is the best on all three metrics
  - Our approach: Maximize for accuracy
- **<u>Final test</u>**: Blind evaluation by experts
  - Top two pipelines based on accuracy





Expert evaluations corroborate the automated metrics

Results

➢ **<u>Final pipeline</u>**: Maximizing for accuracy

The Winning Build

Model: gpt-5-mini-2025-08-07
Retrieval Strategy: OpenAI Keyword
Retrieval Depth: top-k = 7

Accuracy	Latency	Cost
86.66%	**10.04 sec (avg.)**	**$0.005 / query (avg.)**

MIAMI UNIVERSITY

Demonstration